

User interfaces
Desktop GUI (PySide6/Qt) • Chat assistant • Mind-map & charts • CLI

Orchestration
Workflow service • Data models

Ingestion
IAR .map parser • Excel loader • Input detection

Classification
Layer classifier: application / middleware / low-level / system-runtime

Comparison core (deterministic)
Benchmark engine • Multi-signal mapping engine • Acceptance policy
Global & per-layer compare • Recommendations & root-cause

Export
Excel writer • Scoped CSV/JSON • PDF report • GUI payload

Optional AI assistance (default off)
LLM map provider • confident pass • layer classifier • web
ingestion • firmware advisor • AI narrative

optional

Knowledge & persistence (SQLite)
Vendor registry • knowledge store •
feedback / learning • run history